

# **Title:** AI-Powered Virtual Try-On & Fashion Recommendation System

# PROBLEM STATEMENT

- The current fashion e-commerce landscape is plagued by high return rates, often exceeding 30%, primarily due to "size uncertainty" and "visual mismatch"—where the customer cannot accurately judge how a garment will fit or look on their specific body type. While traditional shopping allows for physical trials, online consumers are forced to rely on static model images that rarely reflect their own proportions. This creates a significant gap between consumer expectation and reality, leading to increased logistics costs for retailers and a frustrating user experience.

# Research & Comparative Study

- **Warping Methods:** Evaluated TPS (Geometric) vs. VITON-HD .
- **Recommendation Engine:** Studied Content-Based vs. Hybrid filtering.
- **Datasets:** used VITON-HD/DeepFashion for training.
- **2D Virtual Tryon :** Studied different VTON Models.
- **AR Technologies:** Evaluated different AR technologies

# AI Workflow

1. User uploads image
2. Body detection
3. Pose estimation
4. Body segmentation
5. Body measurement analysis
6. Garment wrapping
7. Virtual Tryon
8. Fashion recommendation
9. Output

# Datasets Collected

- Fashion & virtual try-on datasets
- Deepfashion
- VITON-HD
- ▶ Used for
  - Pose detection
  - Virtual try-on research
  - Model testing

# 2D Virtual Try-On Findings

- **Workflow :**
- User Image → Pose Detection → Body Segmentation → Garment Alignment → Try-On Output
- **Technologies Explored :**
- OpenCV
- MediaPipe
- Human Parsing Models
- HR-VITON Concepts
- **Key Findings:**
- Pose Detection
- Segmentation
- Alignment
- Preprocessing

# 3D Virtual Try-On Findings

- RESEARCH AREAS
- 3D Avatar Generation
- Body Mesh Reconstruction
- Cloth Simulation
- **TOOLS EXPLORED:**
- Blender
- Unity
- <http://Three.js>
- SMPL Models
- **Key Findings:**
- Realistic Visualization
- Interactive Experience
- Real-Time Fitting
- Higher Computational Requirements
- Collaborative Filtering

# Challenges Faced and Solutions

| CHALLENGES             | SOLUTIONS                         |
|------------------------|-----------------------------------|
| Cloth alignment issues | Improved by garment wrapping      |
| Pose detection issues  | Improved landmark detection       |
| Low accuracy           | Tried with high resolution images |
| Occlusion handling     | Checked landmark visibility score |

# Conclusions

- - Implemented real-time human pose estimation using MediaPipe for virtual try-on application.
- - Studied 2D and 3D Virtual Try-On systems including VITON-HD, Stable VITON, and IDM-VTON.
- - Developed an understanding of end-to-end AI fashion workflows from pose detection to garment alignment.